PROJECT NAME: **Vaccine Integrity Monitoring System (VIMS)**

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TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	IoT Data Acquisition & Transmission
2	Real-Time Monitoring Platform & Database
3	AI-Powered Predictive Analytics Engine

SOFTWARE TOOLS FOR TASK 1: IoT Data Acquisition & Transmission

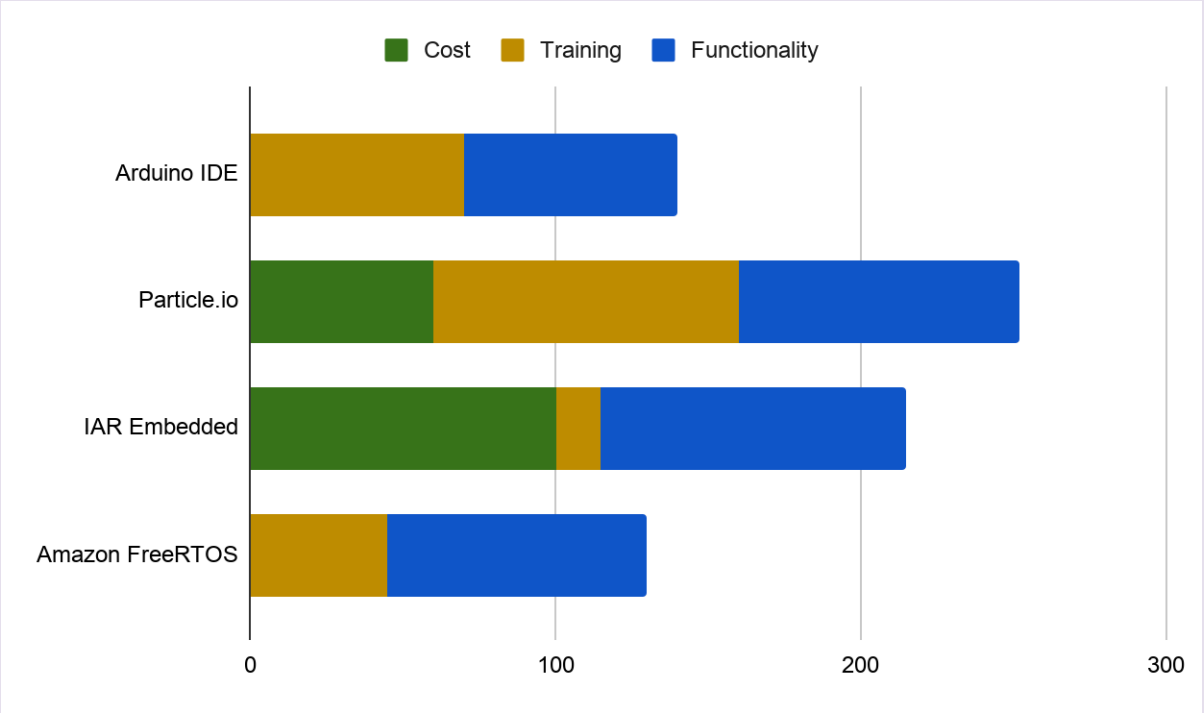
Tool Cost/Training/Functionality Data

Tool	Arduino IDE (C++)	Particle.io (PaaS)	IAR Embedded Workbench	Amazon FreeRTOS
Cost	$0	$1,800	$3,000	$0
Training Days	5 Days	2 Days	12 Days	8 Days
Functionality	70	90	98	85

Normalized Cost/Training/Functionality Data (Higher is better)

Tool	Arduino IDE (C++)	Particle.io (PaaS)	IAR Embedded Workbench	Amazon FreeRTOS
Cost	**0**	60	100	**0**
Training Days	70	**100**	15	45
Functionality	70	92	**100**	85

Normalized Tool Graph



Which tool has been selected? Why?

Selected Tool: Amazon FreeRTOS

We chose Amazon FreeRTOS because it is a professional-grade system built for security and reliability. Since we are handling medical data, its built-in encryption would save the team weeks of development time. It also ensures we meet the 98% transmission goal by automatically reconnecting within 30 seconds if the signal drops during transit.

SOFTWARE TOOLS FOR TASK 2: Real-Time Monitoring Platform & Database

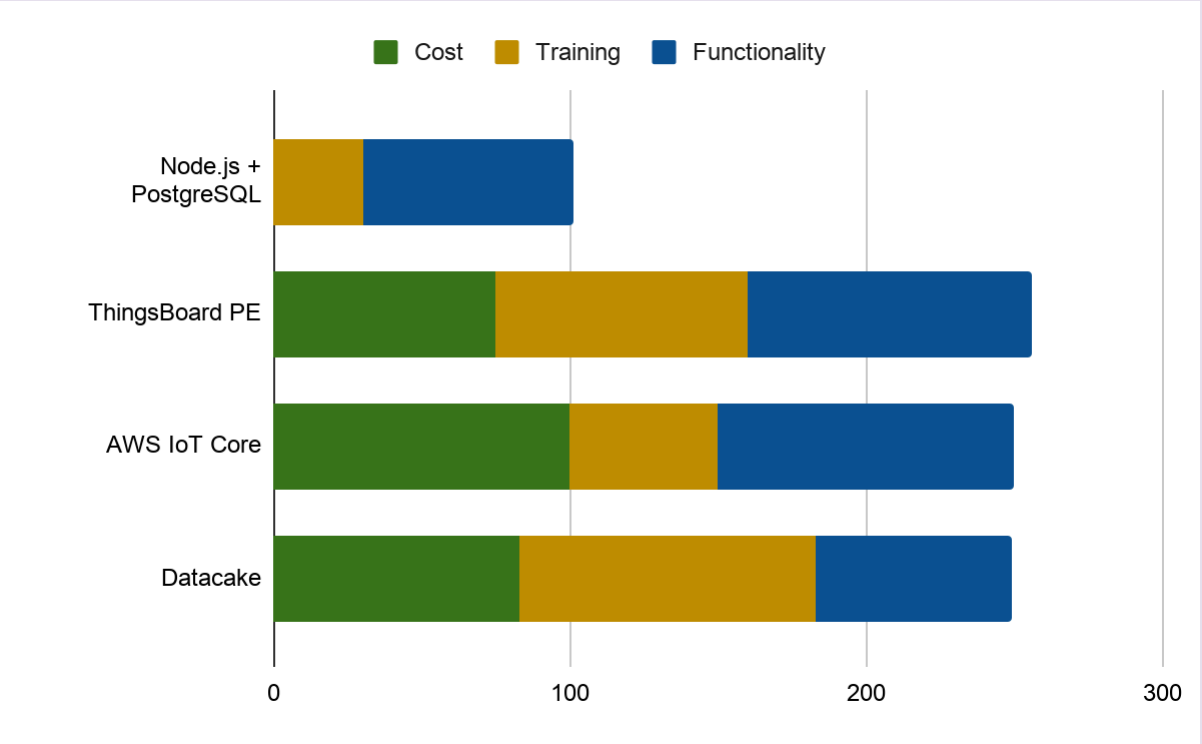
Tool Cost/Training/Functionality Data

Tool	Node.js + PostgreSQL	ThingsBoard PE	AWS IoT Core	Datacake
Cost	$0	$900	$1,200	$1,000
Training Days	15 Days	5 Days	10 Days	1 Day
Functionality	70	94	98	65

Normalized Cost/Training/Functionality Data

Tool	Node.js + PostgreSQL	ThingsBoard PE	AWS IoT Core	Datacake
Cost	**0**	75	100	83
Training Days	30	85	50	**100**
Functionality	71	96	**100**	66

Normalized Tool Graph



Which tool has been selected? Why?

Selected Tool: ThingsBoard PE

ThingsBoard PE is the optimal choice as it offers the best balance across all criteria, providing near-maximum functionality with a significantly lower learning curve than AWS. It is more cost-effective than AWS and offers a more professional, "ready-to-use" monitoring platform than a manual Node.js setup. Ultimately, it provides the perfect middle ground between the complexity of custom coding and the functional limitations of simpler platforms like Datacake.

SOFTWARE TOOLS FOR TASK 3: AI-Powered Predictive Analytics Engine

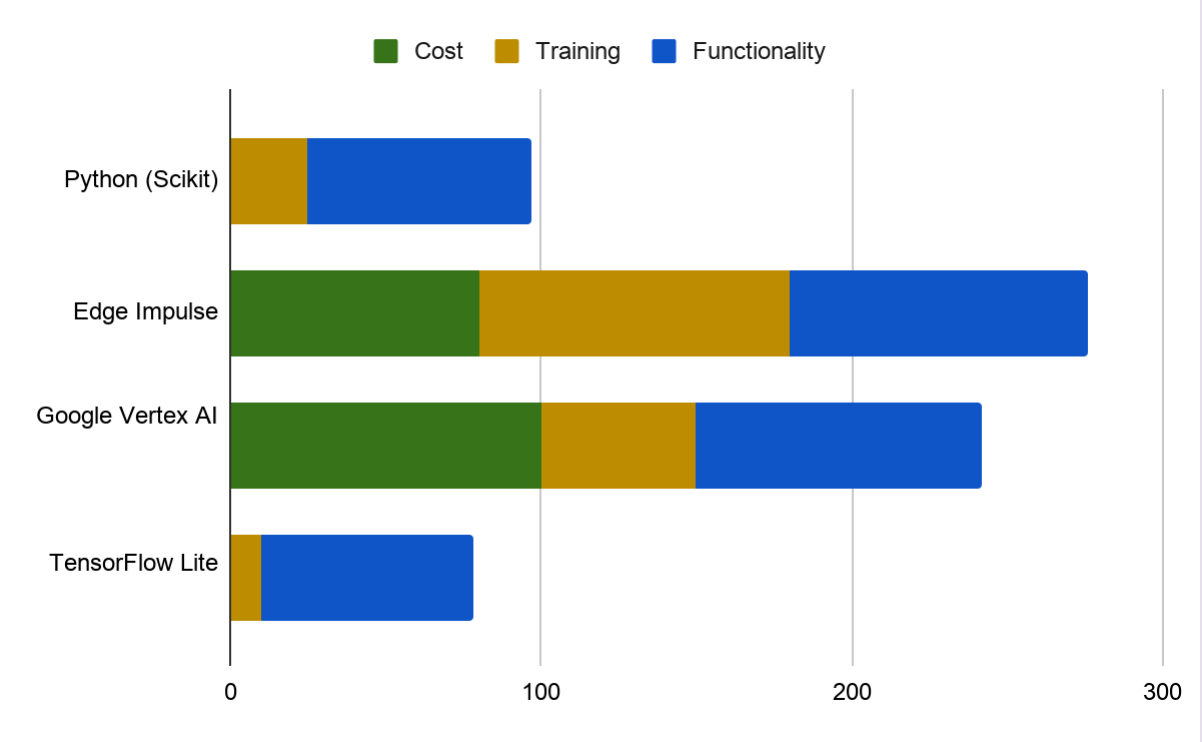
Tool Cost/Training/Functionality Data

Tool	Python (Scikit)	Edge Impulse	Google Vertex AI	TensorFlow Lite
Cost	$0	$1,200	$1,500	$0
Training Days	12 Days	3 Days	7 Days	20 Days
Functionality	72	96	92	68

Normalized Cost/Training/Functionality Data

Tool	Python (Scikit)	Edge Impulse	Google Vertex AI	TensorFlow Lite
Cost	**0**	80	100	**0**
Training Days	25	**100**	50	10
Functionality	72	**96**	92	68

Normalized Tool Graph



Which tool has been selected? Why? **Selected Tool: Edge Impulse** **Edge Impulse is the best choice because it provides the highest performance while being very easy to use. Although it has a certain cost, its technical capabilities and extremely short training time make it much more efficient than the free alternatives. While tools like Python or TensorFlow Lite require expert coding knowledge and a lot of time to learn, Edge Impulse allows the team to develop powerful AI models quickly. It offers a professional-grade platform that delivers better results in less time compared to other complex or lower-functioning options.**